

Calculating rates 1A

1

Use these words to complete the following statements:
different, first, one, quantity, written, per.

A **rate** compares two _____ things.

A rate is usually expressed by writing
how many of the _____ quantity correspond
to _____ of the second _____.

Rates are normally _____ using
the word _____ or the symbol /.

2

Adrian types 440 words in 8 minutes.
What is his typing rate in **words per minute**?

Adrian types 440 words in 8 minutes.
Divide: _____.

He types _____ words in 1 minute.

Adrian's typing rate is _____ words
per minute.

3

Maree scored 42 runs in 10 overs
of cricket. What is her scoring rate
in runs per over?

Divide: _____.
She scored _____ runs in 1 over.

Maree's scoring rate is _____ runs
per over.



4

A water pump fills a tank in 8 hours.
The tank contains 2 500 litres when full.
At what rate is the pump working?
Answer in L/h.

It pumps 2 500 L in 8 hours.

Divide: _____

The rate is: _____

5

Jane delivers 690 bottles of milk in a 3 hour period.
What is her rate per hour?

She delivers 690 bottles in 3 hours.

Divide: _____

The rate is: _____

6

A motorist bought 40 litres of petrol. It cost \$51.60.
What was the cost of petrol per litre?

The petrol cost \$51.60 for 40 litres.

Divide: _____

The rate is: _____



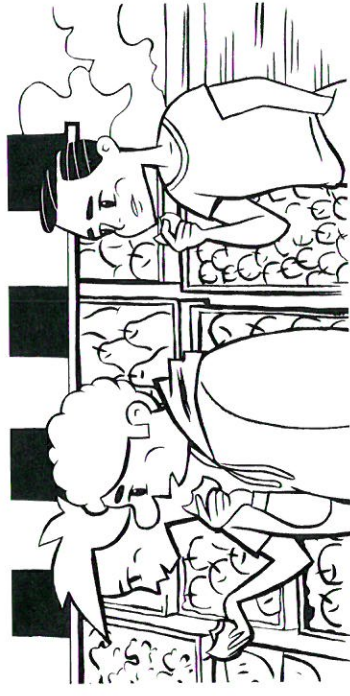
Calculating rates 1B

RATES

UNIT 1
TOPIC 1

7 Calculate the rate of pay per hour.

- A** \$95 for 10 hours _____
- B** \$162 for 9 hours _____
- C** \$105 for 7 hours _____
- D** \$180 for 15 hours _____
- E** \$278.40 for 20 hours _____



8 Calculate the cost per kilogram.

- A** Oranges cost \$3.98 for 2 kg _____
- B** Apples cost \$7.95 for 3 kg _____
- C** Bananas cost \$7.70 for 2 kg _____
- D** Tomatoes cost \$10.35 for 3 kg _____
- E** Potatoes cost \$11.75 for 5 kg _____

9 Michaela scored 870 runs in 10 innings of cricket.
What was her average scoring rate of runs per innings?

10 A tiler lays 56 square metres of floor tiles in 7 hours.
What is his average rate per hour?

11 The tide rose 42 cm in 6 hours.
What was the average rate of the tide rising?

12 The local football team kicked 168 goals in 28 matches.
What is their average scoring rate of goals per match?

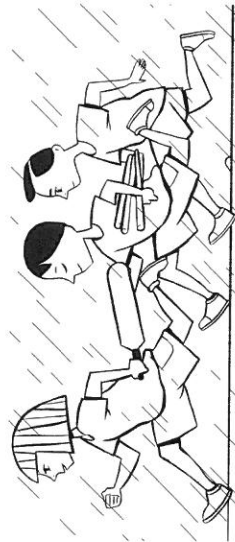
13 A bricklayer lays 460 bricks in 5 hours.
This is an average rate of _____ bricks per hour.

14 A tree grows 156 cm in 3 years.
This is an average growth rate of _____ cm per year.



Calculating rates 2A

- 1** Calculate these rates.
- A** 25 litres in 5 minutes is a rate of ____ L/min.
- B** 150 kilometres in 3 hours is a rate of ____ km/h.
- C** 100 metres in 10 seconds is a rate of ____ m/s.
- D** \$81 in 3 hours is a rate of \$ ____ per hour.
- E** \$630 in 5 days is a rate of \$ ____ per day.
- F** \$840 in 14 weeks is a rate of \$ ____ per week.
- G** 15.6 cm in 6 weeks is a rate of ____ cm/week.
- H** \$345 for 3 nights' accommodation is a rate of \$ ____ per night.
- I** \$37.50 for 3 metres of material is a rate of \$ ____ per metre.
- J** 400 litres in 16 hours is a rate of ____ L/h.
- K** 50 runs for 4 wickets is a rate of ____ runs per wicket.
- L** 144 mm of rain in 6 hours is a rate of ____ mm/h.



- 2** Select the units that may be used to express the following rates by writing A to F in each box.

- A** typing rate **B** cost of potatoes
- C** pulse rate **D** population growth
- E** speed of a car **F** growth of a plant

☐	persons per year	☐	millimetres per week
☐	beats per minute	☐	words per minute
☐	dollars per kilogram	☐	kilometres per hour

- 3** During a netball season, a team used four goalers. The table shows the number of goals each player scored.

Player's name	Goals scored	Number of attempts	Success rate
Pat	19	25	
Helen	32	40	
Malene	21	30	
Peta	6	12	

- A** Calculate the success rate for each player. (Goals scored ÷ number of attempts).

- B** Which player had the highest success rate? ____

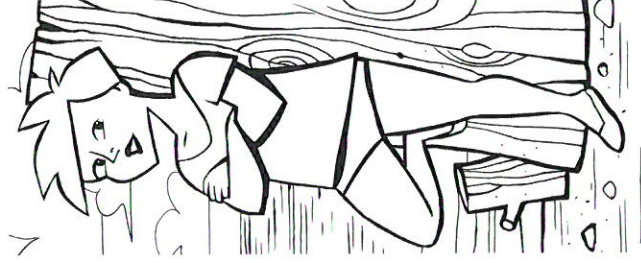
4 Shaun is paid \$340.80 for working 30 hours.
What is his hourly rate of pay?
(Answer to the nearest cent.)

5 A farmer produces 24 000 kg of potatoes
on a farm measuring 20 hectares.
What is the rate of production in kg/ha?

6 A water pump fills a 1.5 kilolitre tank in 8 hours.
At what rate is a pump working?
(Answer in litres per hour.)

7 A tree grows 312 mm in height
in 13 weeks.
Calculate its growth rate
in mm/week.

8 Lisa's heart beats 396 times
in 6 minutes.
Calculate her heartbeat rate.



The *density* of a substance can be expressed in grams per cubic centimetre (g/cm^3).
It is calculated by dividing the mass by its volume.

- 9** Calculate the density of a piece of metal
with mass 110 g and volume 20 cm^3 .

- 10** A piece of wood has mass 450 g and volume 500 cm^3 .
Calculate its density.

11 A piece of gold has a volume of 2.5 cm^3 .
Its mass is 30 grams. Calculate its density.

The petrol consumption of a car is usually expressed
in litres used per 100 km ($\text{L}/100 \text{ km}$).

12 Calculate the petrol consumption
in $\text{L}/100 \text{ km}$ of a car that uses:

- A** 20 L for 200 km. _____
- B** 17 L for 200 km. _____
- C** 27 L for 300 km. _____
- D** 42 L for 500 km. _____
- E** 60.8 L for 800 km. _____